$3d^7$

August 25, 2026

1 $\left|3d_{yz}^2 3d_{z^2}^2 3d_{xz}^\alpha 3d_{xy}^\alpha 3d_{x^2-y^2}^\alpha\right\rangle$

Here are the CSFs:

$$|1,3/2,3/2\rangle = \left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha\right\rangle$$

$$|2,3/2,3/2\rangle = \left|3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha\right\rangle$$

$$|3,3/2,3/2\rangle = \left|3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\alpha\right\rangle$$

$$|4,3/2,3/2\rangle = \left|3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^2\right\rangle$$

$$|5,3/2,3/2\rangle = \left|3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha\right\rangle$$

$$|6,3/2,3/2\rangle = \left|3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\alpha\right\rangle$$

$$|7,3/2,3/2\rangle = \left|3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^2\right\rangle$$

$$|8,3/2,3/2\rangle = \left|3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\alpha\right\rangle$$

$$|9,3/2,3/2\rangle = \left|3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^2\right\rangle$$

$$|10,3/2,3/2\rangle = \left|3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle$$

$$|11,3/2,1/2\rangle = \frac{1}{\sqrt{3}}\left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha\right\rangle$$

$$
\begin{aligned}
|12,3/2,1/2\rangle =& \frac{1}{\sqrt{3}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle \\
&+ \frac{1}{\sqrt{3}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle
\end{aligned}
$$

$$
\begin{aligned}
|13,3/2,1/2\rangle =& \frac{1}{\sqrt{3}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle \\
&+ \frac{1}{\sqrt{3}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle
\end{aligned}
$$

$$
\begin{aligned}
|14,3/2,1/2\rangle =& \frac{1}{\sqrt{3}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle \\
&+ \frac{1}{\sqrt{3}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle
\end{aligned}
$$

$$
\begin{aligned}
|15,3/2,1/2\rangle =& \frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle \\
&+ \frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle
\end{aligned}
$$

$$
\begin{aligned}
|16,3/2,1/2\rangle =& \frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle \\
&+ \frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle
\end{aligned}
$$

$$
\begin{aligned}
|17,3/2,1/2\rangle =& \frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle \\
&+ \frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle
\end{aligned}
$$

$$
\begin{aligned}
|18,3/2,1/2\rangle =& \frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle \\
&+ \frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle
\end{aligned}
$$

$$
\begin{aligned}
|19,3/2,1/2\rangle =& \frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle \\
&+ \frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle
\end{aligned}
$$

$$
\begin{aligned}
|20,3/2,1/2\rangle =& \frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle \\
&+ \frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle
\end{aligned}
$$

$$\begin{aligned}
|21, 3/2, -1/2\rangle =& \frac{1}{\sqrt{3}} \left| 3d_{xz}^{2} 3d_{yz}^{2} 3d_{xy}^{\alpha} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{\beta} \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^{2} 3d_{yz}^{2} 3d_{xy}^{\beta} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{\beta} \right\rangle \\
&+ \frac{1}{\sqrt{3}} \left| 3d_{xz}^{2} 3d_{yz}^{2} 3d_{xy}^{\beta} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{\alpha} \right\rangle \\
|22, 3/2, -1/2\rangle =& \frac{1}{\sqrt{3}} \left| 3d_{xz}^{2} 3d_{yz}^{\alpha} 3d_{xy}^{2} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{\beta} \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^{2} 3d_{yz}^{\beta} 3d_{xy}^{2} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{\beta} \right\rangle \\
&+ \frac{1}{\sqrt{3}} \left| 3d_{xz}^{2} 3d_{yz}^{\beta} 3d_{xy}^{2} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{\alpha} \right\rangle \\
|23, 3/2, -1/2\rangle =& \frac{1}{\sqrt{3}} \left| 3d_{xz}^{2} 3d_{yz}^{\alpha} 3d_{xy}^{\beta} 3d_{z^2}^{2} 3d_{x^2-y^2}^{\beta} \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^{2} 3d_{yz}^{\beta} 3d_{xy}^{\alpha} 3d_{z^2}^{2} 3d_{x^2-y^2}^{\beta} \right\rangle \\
&+ \frac{1}{\sqrt{3}} \left| 3d_{xz}^{2} 3d_{yz}^{\beta} 3d_{xy}^{\beta} 3d_{z^2}^{2} 3d_{x^2-y^2}^{\alpha} \right\rangle \\
|24, 3/2, -1/2\rangle =& \frac{1}{\sqrt{3}} \left| 3d_{xz}^{2} 3d_{yz}^{\alpha} 3d_{xy}^{\beta} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{2} \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^{2} 3d_{yz}^{\beta} 3d_{xy}^{\alpha} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{2} \right\rangle \\
&+ \frac{1}{\sqrt{3}} \left| 3d_{xz}^{2} 3d_{yz}^{\beta} 3d_{xy}^{\beta} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{2} \right\rangle \\
|25, 3/2, -1/2\rangle =& \frac{1}{\sqrt{3}} \left| 3d_{xz}^{\alpha} 3d_{yz}^{2} 3d_{xy}^{2} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{\beta} \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^{\beta} 3d_{yz}^{2} 3d_{xy}^{2} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{\beta} \right\rangle \\
&+ \frac{1}{\sqrt{3}} \left| 3d_{xz}^{\beta} 3d_{yz}^{2} 3d_{xy}^{2} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{\alpha} \right\rangle \\
|26, 3/2, -1/2\rangle =& \frac{1}{\sqrt{3}} \left| 3d_{xz}^{\alpha} 3d_{yz}^{2} 3d_{xy}^{\beta} 3d_{z^2}^{2} 3d_{x^2-y^2}^{\beta} \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^{\beta} 3d_{yz}^{2} 3d_{xy}^{\alpha} 3d_{z^2}^{2} 3d_{x^2-y^2}^{\beta} \right\rangle \\
&+ \frac{1}{\sqrt{3}} \left| 3d_{xz}^{\beta} 3d_{yz}^{2} 3d_{xy}^{\beta} 3d_{z^2}^{2} 3d_{x^2-y^2}^{\alpha} \right\rangle \\
|27, 3/2, -1/2\rangle =& \frac{1}{\sqrt{3}} \left| 3d_{xz}^{\alpha} 3d_{yz}^{2} 3d_{xy}^{\beta} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{2} \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^{\beta} 3d_{yz}^{2} 3d_{xy}^{\alpha} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{2} \right\rangle \\
&+ \frac{1}{\sqrt{3}} \left| 3d_{xz}^{\beta} 3d_{yz}^{2} 3d_{xy}^{\beta} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{2} \right\rangle \\
|28, 3/2, -1/2\rangle =& \frac{1}{\sqrt{3}} \left| 3d_{xz}^{\alpha} 3d_{yz}^{\beta} 3d_{xy}^{2} 3d_{z^2}^{2} 3d_{x^2-y^2}^{\beta} \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^{\beta} 3d_{yz}^{\alpha} 3d_{xy}^{2} 3d_{z^2}^{2} 3d_{x^2-y^2}^{\beta} \right\rangle \\
&+ \frac{1}{\sqrt{3}} \left| 3d_{xz}^{\beta} 3d_{yz}^{\beta} 3d_{xy}^{2} 3d_{z^2}^{2} 3d_{x^2-y^2}^{\alpha} \right\rangle \\
|29, 3/2, -1/2\rangle =& \frac{1}{\sqrt{3}} \left| 3d_{xz}^{\alpha} 3d_{yz}^{\beta} 3d_{xy}^{2} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{2} \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^{\beta} 3d_{yz}^{\alpha} 3d_{xy}^{2} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{2} \right\rangle \\
&+ \frac{1}{\sqrt{3}} \left| 3d_{xz}^{\beta} 3d_{yz}^{\beta} 3d_{xy}^{2} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{2} \right\rangle
\end{aligned}$$

$$|30,3/2,-1/2\rangle = \frac{1}{\sqrt{3}}\left|3d^{\alpha}_{xz}3d^{\beta}_{yz}3d^{\beta}_{xy}3d^{2}_{z^2}3d^{2}_{x^2-y^2}\right\rangle + \frac{1}{\sqrt{3}}\left|3d^{\beta}_{xz}3d^{\alpha}_{yz}3d^{\beta}_{xy}3d^{2}_{z^2}3d^{2}_{x^2-y^2}\right\rangle + \frac{1}{\sqrt{3}}\left|3d^{\beta}_{xz}3d^{\beta}_{yz}3d^{\alpha}_{xy}3d^{2}_{z^2}3d^{2}_{x^2-y^2}\right\rangle$$

$$|31,3/2,-3/2\rangle = \left|3d^{2}_{xz}3d^{2}_{yz}3d^{\beta}_{xy}3d^{\beta}_{z^2}3d^{\beta}_{x^2-y^2}\right\rangle$$

$$|32,3/2,-3/2\rangle = \left|3d^{2}_{xz}3d^{\beta}_{yz}3d^{2}_{xy}3d^{\beta}_{z^2}3d^{\beta}_{x^2-y^2}\right\rangle$$

$$|33,3/2,-3/2\rangle = \left|3d^{2}_{xz}3d^{\beta}_{yz}3d^{\beta}_{xy}3d^{2}_{z^2}3d^{\beta}_{x^2-y^2}\right\rangle$$

$$|34,3/2,-3/2\rangle = \left|3d^{2}_{xz}3d^{\beta}_{yz}3d^{\beta}_{xy}3d^{\beta}_{z^2}3d^{2}_{x^2-y^2}\right\rangle$$

$$|35,3/2,-3/2\rangle = \left|3d^{\beta}_{xz}3d^{2}_{yz}3d^{2}_{xy}3d^{\beta}_{z^2}3d^{\beta}_{x^2-y^2}\right\rangle$$

$$|36,3/2,-3/2\rangle = \left|3d^{\beta}_{xz}3d^{2}_{yz}3d^{\beta}_{xy}3d^{2}_{z^2}3d^{\beta}_{x^2-y^2}\right\rangle$$

$$|37,3/2,-3/2\rangle = \left|3d^{\beta}_{xz}3d^{2}_{yz}3d^{\beta}_{xy}3d^{\beta}_{z^2}3d^{2}_{x^2-y^2}\right\rangle$$

$$|38,3/2,-3/2\rangle = \left|3d^{\beta}_{xz}3d^{\beta}_{yz}3d^{2}_{xy}3d^{2}_{z^2}3d^{\beta}_{x^2-y^2}\right\rangle$$

$$|39,3/2,-3/2\rangle = \left|3d^{\beta}_{xz}3d^{\beta}_{yz}3d^{2}_{xy}3d^{\beta}_{z^2}3d^{2}_{x^2-y^2}\right\rangle$$

$$|40,3/2,-3/2\rangle = \left|3d^{\beta}_{xz}3d^{\beta}_{yz}3d^{\beta}_{xy}3d^{2}_{z^2}3d^{2}_{x^2-y^2}\right\rangle$$

$$|41,1/2,1/2\rangle = +\frac{1}{\sqrt{2}}\left|3d^{2}_{xz}3d^{2}_{yz}3d^{\alpha}_{xy}3d^{\beta}_{z^2}3d^{\alpha}_{x^2-y^2}\right\rangle - \frac{1}{\sqrt{2}}\left|3d^{2}_{xz}3d^{2}_{yz}3d^{\beta}_{xy}3d^{\alpha}_{z^2}3d^{\alpha}_{x^2-y^2}\right\rangle$$

$$|42,1/2,1/2\rangle = +\frac{1}{\sqrt{2}}\left|3d^{2}_{xz}3d^{\alpha}_{yz}3d^{2}_{xy}3d^{\beta}_{z^2}3d^{\alpha}_{x^2-y^2}\right\rangle - \frac{1}{\sqrt{2}}\left|3d^{2}_{xz}3d^{\beta}_{yz}3d^{2}_{xy}3d^{\alpha}_{z^2}3d^{\alpha}_{x^2-y^2}\right\rangle$$

$$|43,1/2,1/2\rangle = +\frac{1}{\sqrt{2}}\left|3d^{2}_{xz}3d^{\alpha}_{yz}3d^{\beta}_{xy}3d^{2}_{z^2}3d^{\alpha}_{x^2-y^2}\right\rangle - \frac{1}{\sqrt{2}}\left|3d^{2}_{xz}3d^{\beta}_{yz}3d^{\alpha}_{xy}3d^{2}_{z^2}3d^{\alpha}_{x^2-y^2}\right\rangle$$

$$|44,1/2,1/2\rangle = +\frac{1}{\sqrt{2}}\left|3d^{2}_{xz}3d^{\alpha}_{yz}3d^{\beta}_{xy}3d^{\alpha}_{z^2}3d^{2}_{x^2-y^2}\right\rangle - \frac{1}{\sqrt{2}}\left|3d^{2}_{xz}3d^{\beta}_{yz}3d^{\alpha}_{xy}3d^{\alpha}_{z^2}3d^{2}_{x^2-y^2}\right\rangle$$

$$|45,1/2,1/2\rangle = +\frac{1}{\sqrt{2}}\left|3d^{\alpha}_{xz}3d^{2}_{yz}3d^{2}_{xy}3d^{\beta}_{z^2}3d^{\alpha}_{x^2-y^2}\right\rangle - \frac{1}{\sqrt{2}}\left|3d^{\beta}_{xz}3d^{2}_{yz}3d^{2}_{xy}3d^{\alpha}_{z^2}3d^{\alpha}_{x^2-y^2}\right\rangle$$

$$|46,1/2,1/2\rangle = +\frac{1}{\sqrt{2}}\left|3d^{\alpha}_{xz}3d^{2}_{yz}3d^{\beta}_{xy}3d^{2}_{z^2}3d^{\alpha}_{x^2-y^2}\right\rangle - \frac{1}{\sqrt{2}}\left|3d^{\beta}_{xz}3d^{2}_{yz}3d^{\alpha}_{xy}3d^{2}_{z^2}3d^{\alpha}_{x^2-y^2}\right\rangle$$

$$|47,1/2,1/2\rangle = +\frac{1}{\sqrt{2}}\left|3d^{\alpha}_{xz}3d^{2}_{yz}3d^{\beta}_{xy}3d^{\alpha}_{z^2}3d^{2}_{x^2-y^2}\right\rangle - \frac{1}{\sqrt{2}}\left|3d^{\beta}_{xz}3d^{2}_{yz}3d^{\alpha}_{xy}3d^{\alpha}_{z^2}3d^{2}_{x^2-y^2}\right\rangle$$

$$|48, 1/2, 1/2\rangle = +\frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|49, 1/2, 1/2\rangle = +\frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$

$$|50, 1/2, 1/2\rangle = +\frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle$$

$$|51, 1/2, -1/2\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|52, 1/2, -1/2\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|53, 1/2, -1/2\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|54, 1/2, -1/2\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle$$

$$|55, 1/2, -1/2\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|56, 1/2, -1/2\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|57, 1/2, -1/2\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle$$

$$|58, 1/2, -1/2\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|59, 1/2, -1/2\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle$$

$$|60, 1/2, -1/2\rangle = \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle$$

$$\begin{aligned}|61, 1/2, 1/2\rangle =& \sqrt{\frac{2}{3}}\left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle \\ &- \frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle\end{aligned}$$

$$\begin{aligned}|62, 1/2, 1/2\rangle =& \sqrt{\frac{2}{3}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle \\ &- \frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle\end{aligned}$$

$$\begin{aligned}|63,1/2,1/2\rangle =&\sqrt{\frac{2}{3}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle-\frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle\\&-\frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle\end{aligned}$$

$$\begin{aligned}|64,1/2,1/2\rangle =&\sqrt{\frac{2}{3}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle-\frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle\\&-\frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle\end{aligned}$$

$$\begin{aligned}|65,1/2,1/2\rangle =&\sqrt{\frac{2}{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle-\frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle\\&-\frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle\end{aligned}$$

$$\begin{aligned}|66,1/2,1/2\rangle =&\sqrt{\frac{2}{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle-\frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle\\&-\frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle\end{aligned}$$

$$\begin{aligned}|67,1/2,1/2\rangle =&\sqrt{\frac{2}{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle-\frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle\\&-\frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle\end{aligned}$$

$$\begin{aligned}|68,1/2,1/2\rangle =&\sqrt{\frac{2}{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle-\frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle\\&-\frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle\end{aligned}$$

$$\begin{aligned}|69,1/2,1/2\rangle =&\sqrt{\frac{2}{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle-\frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle\\&-\frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle\end{aligned}$$

$$\begin{aligned}|70,1/2,1/2\rangle =&\sqrt{\frac{2}{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle-\frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle\\&-\frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle\end{aligned}$$

$$\begin{aligned}|71,1/2,-1/2\rangle =&\frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle+\frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle\\&-\sqrt{\frac{2}{3}}\left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle\end{aligned}$$

$$|72,1/2,-1/2\rangle = \frac{1}{\sqrt{6}}\left|3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta\right\rangle - \sqrt{\frac{2}{3}}\left|3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha\right\rangle$$

$$|73,1/2,-1/2\rangle = \frac{1}{\sqrt{6}}\left|3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\beta\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\beta\right\rangle - \sqrt{\frac{2}{3}}\left|3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\alpha\right\rangle$$

$$|74,1/2,-1/2\rangle = \frac{1}{\sqrt{6}}\left|3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^2\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^2\right\rangle - \sqrt{\frac{2}{3}}\left|3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^2\right\rangle$$

$$|75,1/2,-1/2\rangle = \frac{1}{\sqrt{6}}\left|3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta\right\rangle - \sqrt{\frac{2}{3}}\left|3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha\right\rangle$$

$$|76,1/2,-1/2\rangle = \frac{1}{\sqrt{6}}\left|3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\beta\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\beta\right\rangle - \sqrt{\frac{2}{3}}\left|3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\alpha\right\rangle$$

$$|77,1/2,-1/2\rangle = \frac{1}{\sqrt{6}}\left|3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^2\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^2\right\rangle - \sqrt{\frac{2}{3}}\left|3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^2\right\rangle$$

$$|78,1/2,-1/2\rangle = \frac{1}{\sqrt{6}}\left|3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\beta\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\beta\right\rangle - \sqrt{\frac{2}{3}}\left|3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\alpha\right\rangle$$

$$|79,1/2,-1/2\rangle = \frac{1}{\sqrt{6}}\left|3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^2\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^2\right\rangle - \sqrt{\frac{2}{3}}\left|3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^2\right\rangle$$

$$|80,1/2,-1/2\rangle = \frac{1}{\sqrt{6}}\left|3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle - \sqrt{\frac{2}{3}}\left|3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle$$

$$|81, 1/2, 1/2\rangle = \left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^0\right\rangle$$

$$|82, 1/2, 1/2\rangle = \left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^\alpha\right\rangle$$

$$|83, 1/2, 1/2\rangle = \left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^0\right\rangle$$

$$|84, 1/2, 1/2\rangle = \left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^2\right\rangle$$

$$|85, 1/2, 1/2\rangle = \left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^\alpha\right\rangle$$

$$|86, 1/2, 1/2\rangle = \left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^2\right\rangle$$

$$|87, 1/2, 1/2\rangle = \left|3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^0\right\rangle$$

$$|88, 1/2, 1/2\rangle = \left|3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^2\right\rangle$$

$$|89, 1/2, 1/2\rangle = \left|3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle$$

$$|90, 1/2, 1/2\rangle = \left|3d_{xz}^2 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\alpha\right\rangle$$

$$|91, 1/2, 1/2\rangle = \left|3d_{xz}^2 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^2\right\rangle$$

$$|92, 1/2, 1/2\rangle = \left|3d_{xz}^2 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle$$

$$|93, 1/2, 1/2\rangle = \left|3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^0\right\rangle$$

$$|94, 1/2, 1/2\rangle = \left|3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^2\right\rangle$$

$$|95, 1/2, 1/2\rangle = \left|3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle$$

$$|96, 1/2, 1/2\rangle = \left|3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle$$

$$|97, 1/2, 1/2\rangle = \left|3d_{xz}^0 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\alpha\right\rangle$$

$$|98, 1/2, 1/2\rangle = \left|3d_{xz}^0 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^2\right\rangle$$

$$|99, 1/2, 1/2\rangle = \left|3d_{xz}^0 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle$$

$$|100, 1/2, 1/2\rangle = \left|3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle$$

$$|101, 1/2, -1/2\rangle = \left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^0\right\rangle$$

$$|102, 1/2, -1/2\rangle = \left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^\beta\right\rangle$$

$$|103, 1/2, -1/2\rangle = \left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^0\right\rangle$$

$$|104, 1/2, -1/2\rangle = \left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^2\right\rangle$$

$$|105, 1/2, -1/2\rangle = \left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^\beta\right\rangle$$

$$|106, 1/2, -1/2\rangle = \left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^2\right\rangle$$

$$|107, 1/2, -1/2\rangle = \left|3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^0\right\rangle$$

$$|108, 1/2, -1/2\rangle = \left|3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^2\right\rangle$$

$$|109, 1/2, -1/2\rangle = \left|3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle$$

$$|110, 1/2, -1/2\rangle = \left|3d_{xz}^2 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\beta\right\rangle$$

$$|111, 1/2, -1/2\rangle = \left|3d_{xz}^2 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^2\right\rangle$$

$$|112, 1/2, -1/2\rangle = \left|3d_{xz}^2 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle$$

$$|113, 1/2, -1/2\rangle = \left|3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^0\right\rangle$$

$$|114, 1/2, -1/2\rangle = \left|3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^2\right\rangle$$

$$|115, 1/2, -1/2\rangle = \left|3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle$$

$$|116, 1/2, -1/2\rangle = \left|3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle$$

$$|117, 1/2, -1/2\rangle = \left|3d_{xz}^0 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\beta\right\rangle$$

$$|118, 1/2, -1/2\rangle = \left|3d_{xz}^0 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^2\right\rangle$$

$$|119, 1/2, -1/2\rangle = \left|3d_{xz}^0 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle$$

$$|120, 1/2, -1/2\rangle = \left|3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle$$

The Ground State CSFs are 6 16 26 36

$\left\langle 6, \frac{3}{2}, \frac{3}{2} \left| L_x \right| 10, \frac{3}{2}, \frac{3}{2} \right\rangle = -I$

Δg_{xx}

$+\frac{2}{3}\zeta\Delta^{-1}_{6-10}$

$\left\langle 6,\frac{3}{2},\frac{3}{2}\left|L_y\right|1,\frac{3}{2},\frac{3}{2}\right\rangle = -\sqrt{3}I$
$\left\langle 6,\frac{3}{2},\frac{3}{2}\left|L_y\right|8,\frac{3}{2},\frac{3}{2}\right\rangle = I$

Δg_{yy}

$+2\zeta\Delta^{-1}_{6-1}$
$+\frac{2}{3}\zeta\Delta^{-1}_{6-8}$

$\left\langle 6,\frac{3}{2},\frac{3}{2}\left|L_z\right|3,\frac{3}{2},\frac{3}{2}\right\rangle = -I$

Δg_{zz}

$+\frac{2}{3}\zeta\Delta^{-1}_{6-3}$

For a multiplicity of 4 we have the following:

$D = \frac{1}{2}\left(\left\langle\frac{3}{2},\frac{3}{2}|\frac{3}{2},\frac{3}{2}\right\rangle - \left\langle\frac{3}{2},\frac{1}{2}|\frac{3}{2},\frac{1}{2}\right\rangle\right)$

$E = \frac{1}{\sqrt{3}}\left\langle\frac{3}{2},-\frac{1}{2}|\frac{3}{2},\frac{3}{2}\right\rangle$

$$\begin{aligned}\left\langle\frac{3}{2},\frac{3}{2}\left|\hat{H}_{eff}\right|\frac{3}{2},\frac{3}{2}\right\rangle = -\frac{4}{9}\zeta^2 (\\ &+\frac{1}{4}\Delta^{-1}_{3}+\frac{1}{4}\Delta^{-1}_{11}+\frac{1}{12}\Delta^{-1}_{18}+\frac{1}{12}\Delta^{-1}_{20}+\frac{3}{8}\Delta^{-1}_{41}+\frac{1}{8}\Delta^{-1}_{48} \\ &+\frac{1}{8}\Delta^{-1}_{50}+\frac{1}{8}\Delta^{-1}_{61}+\frac{1}{24}\Delta^{-1}_{68}+\frac{1}{24}\Delta^{-1}_{70}+\frac{1}{4}\Delta^{-1}_{83}+\frac{1}{4}\Delta^{-1}_{85} \\ &+\frac{1}{4}\Delta^{-1}_{97}+\frac{1}{4}\Delta^{-1}_{99}\ )\end{aligned}$$

$$\left\langle\frac{3}{2},\frac{1}{2}\left|\hat{H}_{eff}\right|\frac{3}{2},\frac{1}{2}\right\rangle = -\frac{4}{9}\zeta^2\,(
+\frac{1}{4}\Delta_{1}^{-1}+\frac{1}{12}\Delta_{8}^{-1}+\frac{1}{12}\Delta_{10}^{-1}+\frac{1}{36}\Delta_{13}^{-1}+\frac{1}{3}\Delta_{21}^{-1}+\frac{1}{9}\Delta_{28}^{-1}
+\frac{1}{9}\Delta_{30}^{-1}+\frac{1}{6}\Delta_{43}^{-1}+\frac{1}{8}\Delta_{51}^{-1}+\frac{1}{24}\Delta_{58}^{-1}+\frac{1}{24}\Delta_{60}^{-1}+\frac{1}{18}\Delta_{63}^{-1}
+\frac{1}{24}\Delta_{71}^{-1}+\frac{1}{72}\Delta_{78}^{-1}+\frac{1}{72}\Delta_{80}^{-1}+\frac{4}{3}\Delta_{93}^{-1}+\frac{4}{3}\Delta_{95}^{-1}+\frac{1}{12}\Delta_{103}^{-1}
+\frac{1}{12}\Delta_{105}^{-1}+\frac{1}{12}\Delta_{117}^{-1}+\frac{1}{12}\Delta_{119}^{-1}\,)$$

$$\left\langle\frac{3}{2},-\frac{1}{2}\left|\hat{H}_{eff}\right|\frac{3}{2},\frac{3}{2}\right\rangle = -\frac{4}{9}\zeta^2\,(
-\frac{1}{\sqrt{12}}\Delta_{11}^{-1}-\frac{1}{3\sqrt{12}}\Delta_{18}^{-1}+\frac{1}{3\sqrt{12}}\Delta_{20}^{-1}+\frac{\sqrt{3}}{8}\Delta_{41}^{-1}+\frac{\sqrt{3}}{24}\Delta_{48}^{-1}
-\frac{\sqrt{3}}{24}\Delta_{50}^{-1}+\frac{\sqrt{3}}{24}\Delta_{61}^{-1}+\frac{\sqrt{3}}{72}\Delta_{68}^{-1}-\frac{\sqrt{3}}{72}\Delta_{70}^{-1}+\frac{\sqrt{3}}{12}\Delta_{83}^{-1}
-\frac{\sqrt{3}}{12}\Delta_{85}^{-1}-\frac{\sqrt{3}}{12}\Delta_{97}^{-1}+\frac{\sqrt{3}}{12}\Delta_{99}^{-1}\,)$$

$$D = \frac{2}{9}\zeta^2\,(
+\frac{1}{4}\Delta_{1}^{-1}-\frac{1}{4}\Delta_{3}^{-1}+\frac{1}{12}\Delta_{8}^{-1}+\frac{1}{12}\Delta_{10}^{-1}-\frac{1}{4}\Delta_{11}^{-1}+\frac{1}{36}\Delta_{13}^{-1}
-\frac{1}{12}\Delta_{18}^{-1}-\frac{1}{12}\Delta_{20}^{-1}+\frac{1}{3}\Delta_{21}^{-1}+\frac{1}{9}\Delta_{28}^{-1}+\frac{1}{9}\Delta_{30}^{-1}-\frac{3}{8}\Delta_{41}^{-1}
+\frac{1}{6}\Delta_{43}^{-1}-\frac{1}{8}\Delta_{48}^{-1}-\frac{1}{8}\Delta_{50}^{-1}+\frac{1}{8}\Delta_{51}^{-1}+\frac{1}{24}\Delta_{58}^{-1}+\frac{1}{24}\Delta_{60}^{-1}
-\frac{1}{8}\Delta_{61}^{-1}+\frac{1}{18}\Delta_{63}^{-1}-\frac{1}{24}\Delta_{68}^{-1}-\frac{1}{24}\Delta_{70}^{-1}+\frac{1}{24}\Delta_{71}^{-1}+\frac{1}{72}\Delta_{78}^{-1}
+\frac{1}{72}\Delta_{80}^{-1}-\frac{1}{4}\Delta_{83}^{-1}-\frac{1}{4}\Delta_{85}^{-1}+\frac{4}{3}\Delta_{93}^{-1}+\frac{4}{3}\Delta_{95}^{-1}-\frac{1}{4}\Delta_{97}^{-1}
-\frac{1}{4}\Delta_{99}^{-1}+\frac{1}{12}\Delta_{103}^{-1}+\frac{1}{12}\Delta_{105}^{-1}+\frac{1}{12}\Delta_{117}^{-1}+\frac{1}{12}\Delta_{119}^{-1}\,)$$

$$
\begin{aligned}
E = & \frac{4}{9\sqrt{3}}\zeta^2 \, (\\
& + \frac{1}{\sqrt{12}}\Delta_{11}^{-1} + \frac{1}{3\sqrt{12}}\Delta_{18}^{-1} - \frac{1}{3\sqrt{12}}\Delta_{20}^{-1} - \frac{\sqrt{3}}{8}\Delta_{41}^{-1} - \frac{\sqrt{3}}{24}\Delta_{48}^{-1} \\
& + \frac{\sqrt{3}}{24}\Delta_{50}^{-1} - \frac{\sqrt{3}}{24}\Delta_{61}^{-1} - \frac{\sqrt{3}}{72}\Delta_{68}^{-1} + \frac{\sqrt{3}}{72}\Delta_{70}^{-1} - \frac{\sqrt{3}}{12}\Delta_{83}^{-1} \\
& + \frac{\sqrt{3}}{12}\Delta_{85}^{-1} + \frac{\sqrt{3}}{12}\Delta_{97}^{-1} - \frac{\sqrt{3}}{12}\Delta_{99}^{-1} \,)
\end{aligned}
$$